

3.6 Component 6 – Physics: Electricity, magnetism and waves

Electricity is used in domestic and industrial situations to supply energy. Electric current is a flow of electrical charge and measured in amps. Direct current (d.c.) is supplied by cells and alternating current (a.c.) is supplied by the mains, but in both cases the size of the current depends on the resistance in the circuit. When a current flows through a coil of wire an electromagnet is formed, which like permanent magnets, can exert a force over a distance.

Electric currents can also be used to produce electromagnetic waves, which have many uses including the transmission of information and the transfer of energy from one place to another.